COURSE SYLLABUS

CSC10001 – FUNDAMENTALS OF PROGRAMMING

1. GENERAL INFORMATION

Course name:	Fundamentals of Programming.
Course name (in Vietnamese):	Cơ sở lập trình
Course ID:	CSC10012
Knowledge block:	Fundamentals
Number of credits:	4
Credit hours for theory:	45
Credit hours for practice:	30
Credit hours for self-study:	90
Prerequisite:	None
Prior-course:	None
Instructors:	

2. COURSE DESCRIPTION

The course is designed to provide students with basic and advanced concepts of programming in C/C+ syntax. Students will learn how to construct a complete C/C++ program by using basic programming structures (variables, conditions, and loops), and compound structures (structs, arrays, and functions) across multiple project files. Students will also learn how to use different types of pointers for dynamic memory management, string manipulations, and binary files. Moreover, students will also learn to implement basic data structures such as linked list, stack, and queue. During the course, students will practice presentations in English and demonstrate team-work skills.

3. COURSE GOALS

At the end of the course, students are able to

ID	Description	Program LOs
G1	Describe and use basic programming concepts in C/C++.	1.2.1
G2	Describe and use advanced programming concepts in C/C++.	1.2.1
G3	Describe and implement basic data structures in C/C++.	1.2.1, 1.3.1
G4	Write well-organized C/C++ programs to solve basic problems.	1.2.1, 1.3.1
G5	Explain and present programming concepts in English.	2.4.3, 2.4.5
G6	Demonstrate team-work and presentation skills.	2.2.1, 2.2.2, 2.3.2

4. COURSE OUTCOMES

CO	Description	I/T/U
G1.1	Describe basic programming concepts.	I, T
G1.2	Manipulate input and output.	I, T, U
G1.3	Use control flow statements.	I, T, U
G1.4	Use 1-D and 2-D arrays.	I, T, U
G2.1	Describe and use different types of pointers in C/C++.	I, T
G2.2	Apply C/C++ pointers to manage dynamic array and string.	I, T
G2.3	Describe and manipulate binary files in C/C++.	I, T, U

G3.1	Describe and implement linked-list.	I, T
G3.2	Describe and implement stack and queue.	I, T
G4.1	Organize C/C++ program in structs and functions across multiple files.	I, T
G5.1	Explain and present programming concepts in English.	U
G6.1	Demonstrate team-work and presentation skills.	U

5. TEACHING PLAN

ID	Topic	Course outcomes	Teaching/Learning Activities (samples)
1	Programming Overview - Programming: concepts, languages, environments. - Algorithm: concepts, notations, complexity. - C/C++ overview: history, program structure.	G1.1	Lecturing, Demonstration HW1
2	Input and Output - Basic units: variables/constants, data-types, expressions. - Input output stream: concepts, stdin/ stdout, formatting. - Text file: file stream, read/write.	G1.2, G5.1	Lecturing, Demonstration QZ1, HW2
3	Control Flow Statements - Conditions: if-else, switch-case. - Loop statements: while, do-while, for.	G1.3	Lecturing, Demonstration QZ2, HW3

4	Function and Struct - Function: concepts, passing arguments. - Struct: concept and usage. - Complex C/C++ program: header file, multiple-file project.	G4.1, G5.1, G6.1	Lecturing, Demonstration, Case-study QZ3, HW4
5	Array - Array 1-D: concept, manipulations. - Array 2-D: concept, manipulations. - String: C-string vs. std::string.	G1.4	Lecturing, Demonstration QZ4, HW5
6	Pointer - Concept, declaration, operators. - Pointer vs. array. - Dynamic memory management.	G2.1	Lecturing, Demonstration QZ5, HW6
7	Advanced Pointers - C-string manipulation. - Pointer of pointer. - Other types of pointers.	G2.2	Lecturing, Demonstration QZ6, HW7
8	Binary File - Binary vs. text file. - Read/write formatted binary file.	G2.3, G5.1, G6.1	Lecturing, Demonstration, Case-study HW8
9	Linked List - Singly linked list: linked list vs. array, declaration, operations, manipulations. - Other types of linked list.	G3.1	Lecturing, Demonstration HW9
10	Stack and Queue - Stack: concept and operations. - Queue: concept and operations.	G3.2, G5.1, G6.1	Lecturing, Q&A, Group discussion HW10

For the practical laboratory work, there are 10 weeks which cover similar topics as it goes in the theory class. Each week, teaching assistants will explain and demonstrate key ideas on the corresponding topic and ask students to do their lab exercises on computers either in the lab or at home. All the lab work submitted will be graded. There would be a final exam for lab work.

6. ASSESSMENTS

ID	Topic	Description	Course outcomes	Ratio (%)
A1	**Assignments**			**30%**
A11	Homework: HW1-HW10	Homework are programming problems assigned at the end of each session, do it at home and submit online.	G1.1, G1.2, G1.3, G1.4, G2.1, G2.2, G2.3, G3.1, G3.2, G4.1	10%
A12	Lab-work: LW1–LW10	Lab-work are programming problems presented and guided during lab session.	G1.1, G1.2, G1.3, G1.4, G2.1, G2.2, G2.3, G3.1, G3.2, G4.1	10%
A13	Quiz: QZ1-QZ6	Quiz are quick tests at the beginning of each session.	G1.2, G1.3, G1.4, G2.1, G2.2, G4.1	10%
A2	**Exams**			**70%**
A21	Lab midterm exam	In-class programming exam on computer		10%
A22	Lab final exam	In-class programming exam		10%

		on computer		
A23	Midterm exam	Closed book exam. Describe the understanding of different topics, analyze & program to solve problems		10%
A24	Final exam	Closed book exam. Describe the understanding of different topics, analyze & program to solve problems		40%

7. RESOURCES

Textbooks

- K.N. King, C Programming: A Modern Approach, 2th Edition, Norton & Company, 2008.
- Nhập môn lập trình, Trần Đan Thư, Nguyễn Thanh Phương, Đinh Bá Tiến, Trần Minh Triết, NXB Khoa Học Kỹ thuật, 2011
- Kỹ thuật lập trình, Trần Đan Thư, Nguyễn Thanh Phương, Đinh Bá Tiến, Trần Minh Triết, NXB Khoa Học Kỹ thuật, 2011.

Others

- Adam Drozdek, Data Structures and Algorithms in C++, 4th Edition, Cengage Learning, 2008.
- The C Programming Language, 2th Edition, Brian W. Kernighan, Dennis M. Ritchie, Prentice Hall, 1988.
- C Programming, Wikibooks, http://en.wikibooks.org/wiki/C_Programming

8. GENERAL REGULATIONS & POLICIES

- All students are responsible for reading and following strictly the regulations and policies of the school and university.
- Students who are absent for more than 3 theory sessions are not allowed to take the final exam.
- For any kinds of cheating and plagiarism, students will be graded 0 for the course. The incident is then submitted to the school and university for further review.
- Students are required to take the final exam, otherwise they cannot earn the credit.
- Students are encouraged to form study groups to discuss course topics. However, individual work must be done and submitted on your own.